ChatGPT

Empirical Rule Calculation

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Source: https://chatgpt.com/c/698c1c29-6048-8327-bd91-c29989b90e98

If the mean life of the corona virus is 180 min and the standard deviation is 20 min. Then, 68 % of all viruses could be active between _____ min and _____ min.

We can solve this step by step using the **empirical rule** (68–95–99.7 rule) for normal distributions:

- **Mean (μ)** = 180 min
- **Standard deviation (σ)** = 20 min

The **empirical rule** says:

- About **68%** of values lie within **1 standard deviation** of the mean.
- That is:

$$\text{Range} = \mu - \sigma \text{ to } \mu + \sigma$$

Plug in the numbers:

$$\mu - \sigma = 180 - 20 = 160$$

$$\mu + \sigma = 180 + 20 = 200$$

✅ So, **68% of all viruses** could be active between **160 min and 200 min**.

If you want, I can also explain **why this works with the normal curve** in very simple terms. Do you want me to?

The scores of the CST8233 exam are normally distributed with a mean of 84% and a standard deviation of 8. John scored 72%. What is the percentage of students who scored better than John?

Enter your answer as a either a decimal value with 5 digits after the decimal, or as a percent score with 2 digits after the decimal.

Let's solve this **step by step**.

We are given:

- Mean $\mu = 84\%$
- Standard deviation $\sigma = 8\%$
- John's score $X = 72\%$

We want **P(score > 72)**.

Step 1: Compute the Z-score

The formula for Z-score is:

$$Z = \frac{X - \mu}{\sigma}$$

$$Z = \frac{72 - 84}{8} = \frac{-12}{8} = -1.5$$

Step 2: Find probability below John's score

From the **standard normal table**, $P(Z < -1.5) \approx 0.0668$.

This means about **6.68%** of students scored **less than John**.

Step 3: Find probability above John's score

$$P(X > 72) = 1 - P(Z < -1.5)$$

$$P(X > 72) = 1 - 0.0668 = 0.9332$$

✅ Step 4: Convert to percentage

$$0.9332 \times 100 \approx 93.32\%$$

Answer:

$$\boxed{93.32\%}$$

If you want, I can also **draw the normal curve** showing John's score and the area above it — it makes this really easy to visualize. Do you want me to do that?